

Lecture 3: Tasks Solutions

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Your Turn #3

Let $X_1, \dots, X_n$ be i.i.d. random variables with a finite expected value $E[X_i] = \mu$ and $\mathbb{V}(X_i) = \sigma^2 < \infty$.

Question 1

Show that $\mathbb{E}[S_n^2] = \mathbb{E}[X^2] - \mathbb{E}[\bar{X}_n^2]$.

Answer:

$$\mathbb{E}[S_n^2] = \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \right] \quad (1)$$

$$= \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n X_i^2 - \frac{1}{n} 2 \sum_{i=1}^n X_i \bar{X}_n + \frac{1}{n} \sum_{i=1}^n \bar{X}_n^2 \right] \quad (2)$$

$$= \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i^2] - \mathbb{E} \left[\frac{1}{n} 2 \bar{X}_n \sum_{i=1}^n X_i \right] + \mathbb{E} [\bar{X}_n^2] \quad (3)$$

$$= \mathbb{E} [X^2] - \mathbb{E} \left[\frac{1}{n} 2 \bar{X}_n n \bar{X}_n \right] + \mathbb{E} [\bar{X}_n^2] \quad (4)$$

$$= \mathbb{E} [X^2] - 2 \mathbb{E} [\bar{X}_n^2] + \mathbb{E} [\bar{X}_n^2] \quad (5)$$

$$= \mathbb{E} [X^2] - \mathbb{E} [\bar{X}_n^2] \quad (6)$$

Question 2

Show that $\mathbb{E}[X^2] = \sigma^2 + \mu^2$. Hint: recall the formula for the variance using expectations.

Answer:

Recall that $\mathbb{V}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$. We therefore have:

$$\mathbb{E}[X^2] = \mathbb{V}(X) + \mathbb{E}[X]^2 \quad (7)$$

$$= \sigma^2 + \mu^2 \quad (8)$$

Question 3

Show that $\mathbb{E}[\bar{X}_n^2] = \frac{\sigma^2}{n} + \mu^2$. Hint: the formula for $\mathbb{V}(\bar{X}_n)$ will be useful.

Answer:

Using the variance formula and replacing X by $\bar{X}_n$, we have:

$$\mathbb{E}[\bar{X}_n^2] = \mathbb{V}(\bar{X}_n) + \mathbb{E}[\bar{X}_n]^2 \quad (9)$$

$$= \frac{\sigma^2}{n} + \mu^2 \quad (10)$$

Question 4

Combine the above results to show that $\mathbb{E}[S_n^2] \neq \sigma^2$.

Answer:

We can use the results from questions 2 and 3:

$$\mathbb{E}[S_n^2] = \mathbb{E}[X^2] - \mathbb{E}[\bar{X}_n^2] \quad (11)$$

$$= \sigma^2 + \mu^2 - \left(\frac{\sigma^2}{n} + \mu^2\right) \quad (12)$$

$$= \sigma^2 - \frac{\sigma^2}{n} \quad (13)$$

$$= \frac{n-1}{n} \sigma^2 \quad (14)$$

$$\neq \sigma^2 \quad (15)$$

Question 5

Load this [voter data](#). Use R to compute the variance of Pumpernickel voters: (1) manually (i.e., taking the sum of deviations from the mean), and (2) using the `var` command. What do you notice?

Answer:

```
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.5.2

```
voter_data <- read.csv("https://www.dropbox.com/scl/fi/lnz14t3a2cj6kw0l217yo/pumpernickel_voter_data.csv")

voter_data |>
  mutate(deviation_from_mean = pumpernickel_voter - mean(pumpernickel_voter)) |>
  summarise(var_manual = 1/n() * sum(deviation_from_mean^2),
            var_command = var(pumpernickel_voter),
            var_manual_adjusted = n()/(n()-1) * var_manual)
```

```
  var_manual var_command var_manual_adjusted
1    0.1771    0.1772773         0.1772773
```

Question 6

Read the help `sd` (it is simpler than `var`) command's help file to see if you can understand what's going on.

```
?sd
```

Like var this uses denominator $n - 1$.

Your Turn #4

Question 1

Use `pnorm` to confirm the figure on slide 65, i.e., within ± 1 (2) standard deviation of the mean, the standard normal distribution contains 68% (95%) of observations.

Answer:

```
pnorm(q = 1, mean = 0, sd = 1) - pnorm(q = -1, mean = 0, sd = 1)
```

```
[1] 0.6826895
```

```
pnorm(q = 2, mean = 0, sd = 1) - pnorm(q = -2, mean = 0, sd = 1)
```

```
[1] 0.9544997
```

Question 2

A region has household incomes with mean $\mu = 58,000$ euros and standard deviation $\sigma = 15,000$ euros. An economist surveys 100 randomly selected households. What is the probability that the sample mean income is less than 55,000 euros?

Answer:

```
sample_mean = 55000
pop_mean = 58000
pop_sd = 15000
sample_size = 100
mean_se = pop_sd/sqrt(sample_size)

pnorm(sample_mean, mean = pop_mean, sd = mean_se)
```

```
[1] 0.02275013
```

Question 3

A country's true unemployment rate is 6.2%. A labor economist draws a random sample of 1,000 working-age adults. What is the probability that the sample unemployment rate falls between 5% and 7%?

Answer:

```
pop_prop = 0.062
pop_var = 0.062 * (1 - 0.062)
sample_size = 1000
se <- sqrt((pop_prop*(1-pop_prop)) / sample_size)
```

```
lower_bound = 0.05
upper_bound = 0.07

pnorm(upper_bound, mean = pop_prop, sd = se) - pnorm(lower_bound, mean = pop_prop, sd = se)
```

```
[1] 0.7951262
```

Question 4

Monthly consumer spending at retail stores has mean $\mu = 320$ euros and standard deviation $\sigma = 85$ euros. A retail analyst collects data from 50 randomly selected consumers. Below what value does the lowest 10% of sample mean spending fall?

Answer:

```
pop_mean = 320
pop_sd = 85
sample_size = 50
quantile = .1
qnorm(quantile, mean = pop_mean, sd = pop_sd/sqrt(sample_size))
```

```
[1] 304.5947
```